FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE **JEE(Advanced)-2015**

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT –II
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A, B	A, B, D	A, B, C
2.	B, D	A, B, D	B, C
3.	B, D	A, B, D	A, B
4.	B, D	B, D	A, D
5.	A,D	B, D	A, B, C
6.	A,B,D	B, D	A, B
7.	A,C,D	A, B, C	A, B
8.	A	A, B, C, D	B, C, D
9.	B	C	B
10.	A	D	B
11.	A	C	D
12.	C	A	B
13.	C	D	C
14.	C	C	A
15.	A	D	B
16.	D	B	C
17.	A	D	D
18.	C	D	C
19.	C	B	B
20.	B	C	D

Physics

PART – I

SECTION – A

1. $|F| = \frac{dU}{dr} = \frac{ke^2}{r^4}$

$F = \frac{mv^2}{r}$

$\Rightarrow \frac{ke^2}{r^4} = \frac{mv^2}{r}$ ...(1)

and $mvr = \frac{nh}{2\pi}$ ...(2)

From (1) and (2) $r = \frac{ke^2 4\pi^2}{n^2}\frac{m}{n^2} = \frac{Am}{n^2}$

T.E. = $\frac{1}{2}$P.E $= \frac{1}{2} \times \frac{ke^2}{3\left(\frac{Am}{n^2}\right)^3}$

T.E. $\propto n^6$, T.E. $\propto m^{-3}$

2. $\frac{P^2}{\rho} = \text{constant}$

$\Rightarrow \frac{P^2(nRT)}{MP} = \text{constant}$

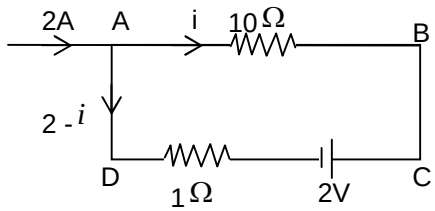
$\Rightarrow PT = \text{constant}$

3. In loop ABCDA

$10i + 2 - (2 - i) = 0$

$i = 0$

Potential difference across S → $(2 - i)1 = 2V$



8. $\omega L/R = \tan\pi/6 \Rightarrow R = 3\Omega$

$\tan\phi = 1/\omega CR = 1$

$\Rightarrow \phi = \pi/4$, leading P.D.

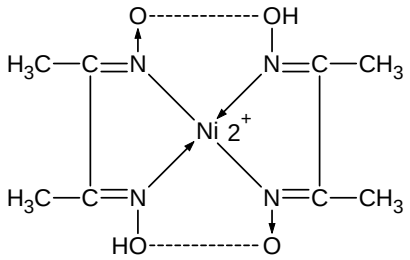
Chemistry

PART – II

SECTION – A

1. $Ba^{2+} \xrightarrow{CrO_4^{2-}} BaCrO_4$
$Pb^{2+} \xrightarrow{CrO_4^{2-}} PbCrO_4$ } Yellow ppt.
$Sr^{2+} \xrightarrow{CrO_4^{2-}} SrCrO_4$

$2Ag^{+} \xrightarrow{CrO_4^{2-}} Ag_2CrO_4$
$Hg_2^{2+} \xrightarrow{CrO_4^{2-}} Hg_2CrO_4$ } Red ppt.

2. 

$[Ni(dmg)_2]$

4. $Zn(s) + 2H^{+}(aq) \rightleftharpoons Zn^{2+}(aq) + H_2$ ....(1)

$$E = E^0_{cell} - \frac{0.059}{n}\log\frac{\left[Zn^{2+}\right][H_2]}{\left[H^{+}\right]^2} \quad .....(2)$$

Equilibrium (1) will shift in forward direction and cell potential E will increase.

5. Rate of zero order reactions is independent of the concentration of reaction
In this equation $K = Ae^{-E_a/RT}$
$K = A$ when $E_a = 0$

$$\log K = \log A - \frac{E_a}{2.303RT}$$

Plot of log K against 1/T is straight line with negative slope.

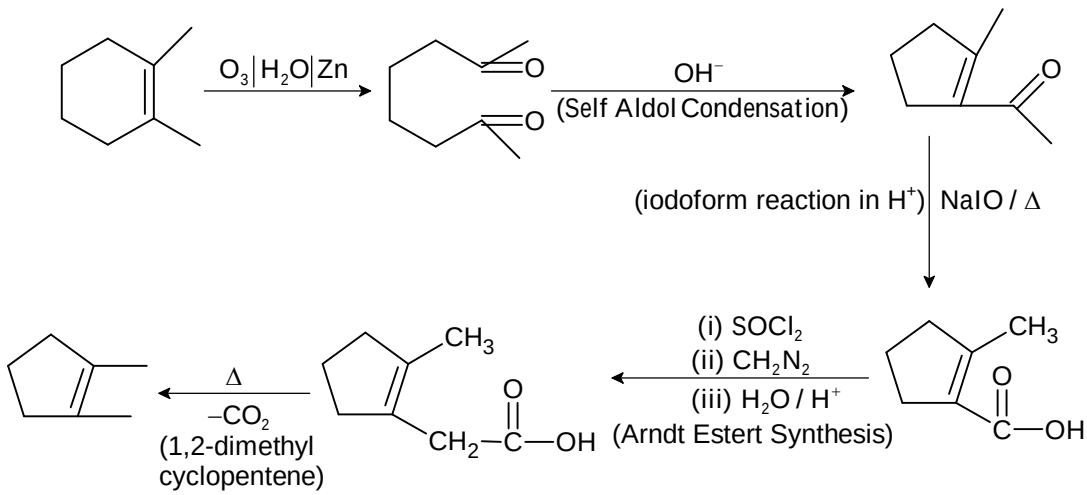
7. At anode: $2Cl^{-} \longrightarrow Cl_2 + 2e^{-}$
At cathode: $Mg^{2+} + 2e^{-} \longrightarrow Mg$

$(\text{No. of gm equivalent of Mg at cathode}) = (\text{No of gm equivalent of } Cl_2 \text{ at Anode})$

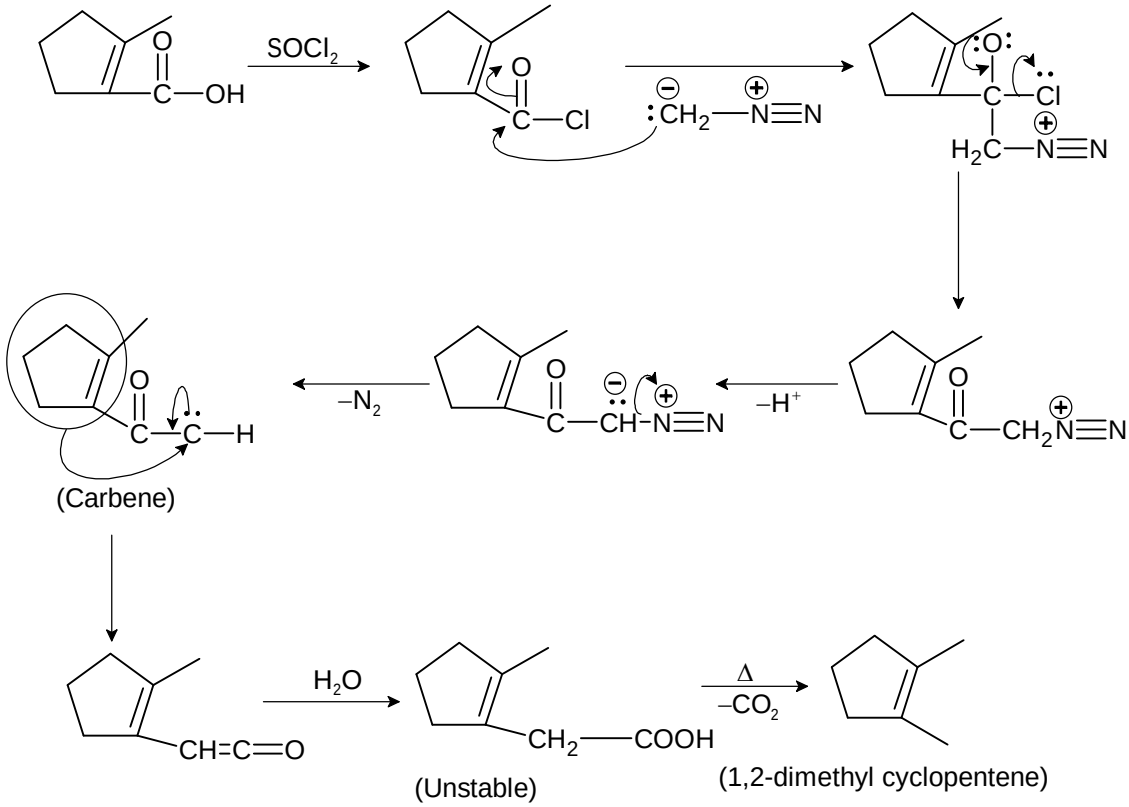
$$\frac{13}{24.3/2} = \frac{W_{Cl_2}}{35.5}$$

$W_{Cl_2} = 37.98\,gm$
or $n_{Cl_2} = 0.535$
V_{Cl_2} = 11.98 lit at NTP

9.

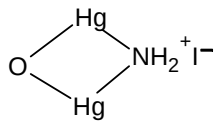


10.



11. $\underset{(A)}{Ba} \xrightarrow[\text{Burning}]{N_2} \underset{(B)}{Ba_3N_2} \xrightarrow{H_2O} Ba(OH)_2 + NH_3\,(D)$

$BaCl_2 \longrightarrow$ green flame

12. $HgI_2 \xrightarrow{KI} K_2HgI_4 \xrightarrow[KOH]{NH_3}$ 

13. Reaction of Mn^{3+} with H_2O is spontaneous.

14. Reduction potential of I_2 is less than Cl_2.

Solution for the Q. No. 15 to 16.

(i) A turns lime water milky so 'A' may be CO_2 or SO_2

(ii) B condenses into liquid turning anhydrous $CuSO_4$ to blue, so 'B' is H_2O.

(iii) Y gives alkaline solution and its solution forms white ppt. Z with $BaCl_2$ and Z on heating with acid gives CO_2 so, Z is $BaCO_3$ and Y is metal carbonate.

(iv) Since, Y and A are formed from 'X' and thus X is metal bicarbonate.

(v) $$\underset{20.02\,g}{X} \longrightarrow \underset{4.4\,g}{A} + \underset{1.8\,g}{B} + \underset{13.8\,g}{Y}$$

(vi) The above data reveal that,

$$\underset{(X)}{2MHCO_3} \longrightarrow \underset{(A)}{CO_2} + \underset{(B)}{H_2O} + \underset{(Y)}{M_2CO_3}$$

4.4 of CO_2 is obtained by 20.02 g of $MHCO_3$

44 g CO_2 is obtained by 200.2 g of $MHCO_3$

Molecular weight of $MHCO_3 = \dfrac{200.2}{2} = 100.1$

$\therefore$ Atomic wt. of $M \approx 39.0$ g/mole

Thus metal is potassium.

17. BF_4^- has sp^3 hybridisation, CH_3^+ is sp^2, F_2O is sp^3 with two lone pairs on O. Also N_2O is linear.

Mathematics

PART – III

SECTION – A

1. $$\frac{(200)!}{\underbrace{2!\,.\,2!\,.\,.....2!}_{100\text{ times}}\,(100)!} = \frac{(200)!}{100!\,.\,2^{100}} = 1.3.5\,.....\,.\,199$$

$$\frac{(200)!}{100!\,.\,2^{100}} = \left(\frac{101}{2}\right)\left(\frac{102}{2}\right).....\left(\frac{200}{2}\right)$$

2. Since, f(–1) = 0 the other root is $-\frac{b+c-a}{a+c-b} \in Q$

Which can be written as $-\frac{\frac{c}{a-b}-1}{\frac{c}{a-b}+1}$

Hence, $\frac{c}{a-b} \in Q$

3. a, b, c are in GP

$\frac{2ab}{a+b} = 12$, $\frac{b+c}{2} = 3$

$\frac{2a\,.\,ar}{a+ar} = 12$, $ar + ar^2 = 6$

$ar = 6(1 + r)$, $ar(1 + r) = 6$

i.e. $6(1 + r)^2 = 6$, i.e. $(1 + r)^2 = 1$

$\therefore$ $1 + r = \pm 1$ i.e. $r = 0$ or $r = -2$

$\therefore$ $r = -2$

If $r = -2$, then $a = 3$

$\therefore$ The GP is 3, –6, 12, –24, 48

Clearly no term of the series is square of an integer AM of a, b, c is = $\frac{3-6+12}{3} = 3$

4. Homogenizing auxiliary circle with normal at P

$$x^2 + y^2 = \frac{a^2(ax\sec\theta - by\,\mathrm{cosec}\theta)^2}{(a^2-b^2)^2}$$

$$\angle QOR = 90^\circ \Rightarrow -1 + \frac{a^4\sec^2\theta}{(a^2-b^2)^2} - 1 + \frac{a^2b^2\,\mathrm{cosec}^2\theta}{(a^2-b^2)^2} = 0$$

$2(a^2 - b^2)^2 = a^4\sec^2\theta + a^2b^2\,\mathrm{cosec}^2\theta$

$a^4 - 5a^2b^2 + 2b^4 = a^4\tan^2\theta + a^2b^2\cot^2\theta \geq 2a^3b$.

5. $$f(x) = \begin{vmatrix} x^2-5x+3 & 2x-5 & 3 \\ 3x^2+x+4 & 6x+1 & 9 \\ 7x^2-6x+9 & 14x-6 & 21 \end{vmatrix}$$

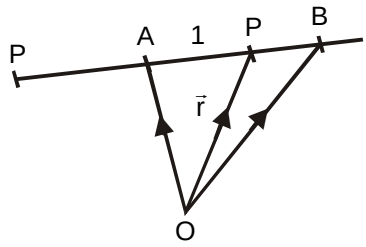
$f'(x) = 0$

$\Rightarrow$ f(x) is a constant polynomial and $f(0) \neq 0$

$\Rightarrow$ $d \neq 0$

6. $(1 + x^2)^2 (1 + x)^n = (1 + 2x^2 + x^4) ({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ...)$
$= {}^nC_0 + {}^nC_1 x + ({}^nC_2 + 2\cdot{}^nC_0) x^2 + ...$
Hence, $A_0 = 1$; $A_1 = {}^nC_1$; $A_2 = {}^nC_2 + 2$ which are in A.P.

7. $\overrightarrow{OP} = \vec{r} = \overrightarrow{OA} \pm \dfrac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$



$= \hat{i} + \hat{j} - 2\hat{k} \pm \dfrac{\overrightarrow{OB} - \overrightarrow{OA}}{|\overrightarrow{OB} - \overrightarrow{OA}|}$

$= \hat{i} + \hat{j} - 2\hat{k} \pm \dfrac{-4\hat{j} + 3\hat{k}}{|-4\hat{j} + 3\hat{k}|}$

$= \hat{i} + \hat{j} - 2\hat{k} \pm \dfrac{1}{5}(-4\hat{j} + 3\hat{k})$

$= \dfrac{1}{5}(5\hat{i} + \hat{j} - 7\hat{k}), \dfrac{1}{5}(5\hat{i} + 9\hat{j} - 13\hat{k})$

8. Put $x = 1/t$ in u or $v \Rightarrow u = v$
Now consider u + v

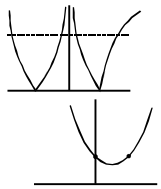
9. $P\left(x \le \dfrac{1}{2}\right) = \int\limits_{x=0}^{1/2} f(x)\,dx = \int\limits_{0}^{1/2} 2x\,dx = \left[x^2\right]_0^{1/2} = \dfrac{1}{4}$

10. Probability that life of a radio tube is less than or equal to 150 hours
$= \int\limits_0^{150} f(x)\,dx = \int\limits_{100}^{150} \dfrac{100}{x^2}\,dx = -\dfrac{100}{x}\Big|_{100}^{150} = -100\left[\dfrac{1}{150} - \dfrac{1}{100}\right] = \dfrac{1}{3}$

$\therefore$ Probability that life is greater than 150 hrs = $\dfrac{2}{3}$

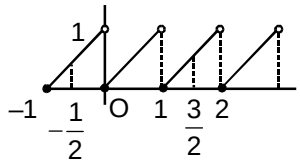
$\therefore$ Probability that none of these have life less than on equal to 150 hrs = $\dfrac{2}{3} \times \dfrac{2}{3} \times \dfrac{2}{3} = \dfrac{8}{27}$ h

11. $y = \left|\ell n|x|\right|$ has graph here an element in co-domain can have 4 pre-images



$y = x^2 \sin x$ is zero for infinitely many values of x
$y = x^3 + 3x + 1$ is M.I. and hence one-one function
$y = x^4 - x + 1$ has graph
$\therefore$ $y = x^4 - x + 1$ is two-one

12. From graph it is clear that any horizontal lines cuts the graph at either one or two points between $\left[-\dfrac{1}{2}, \dfrac{3}{2}\right)$



13. At x = 1, y = 0
$\dfrac{dy}{dx} = 2x\cdot(x^4)^2 - (x^2)^2 = 2 - 1 = 1$
$\therefore$ Equation of tangent $y - 0 = 1\,(x - 1)$, $y = x - 1$

14. $F(x) = \int_1^x e^{t^2/2}(1-t^2)\,dt$

$F'(x) = \left(e^{\frac{x^2}{2}}(1-x^2)\right)^2$

$\therefore F'(1) = e^{1/2}.0 = 0$

15. Let $\sin x = t;\ t \in [-1, 1]$

$\sqrt{5-2t} = 6t-1 \geq 0$ i.e. $t \geq \frac{1}{6}$

So, $t \in \left[\frac{1}{6}, 1\right]$

Now squaring, $\sqrt{5-2t} = (6t-1)$, $t = \frac{1}{2}, -\frac{2}{9}$, $t = -\frac{2}{9}$ not possible $\left(\because t \in \left[\frac{1}{6}, 1\right]\right)$

So, $t = \frac{1}{2}$, $\sin x = \frac{1}{2} = \sin\frac{\pi}{6}$, $x = n\pi + (-1)^n\frac{\pi}{6}$

16. $2\frac{\cos 2x}{\sin 2x} - 3\frac{\cos 3x}{\sin 3x} = \frac{\sin 2x}{\cos 2x}$

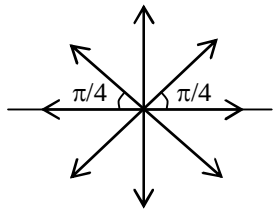
$2\frac{(2\cos^2 x - 1)}{2\sin x\cos x} - 3\frac{4\cos^3 x - 3\cos x}{3\sin x - 4\sin^3 x} = \frac{2\sin x\cos x}{2\cos^2 x - 1}$

Now Let $\cos^2 x = t$ in above i.e. $t \in (0,1) - \left\{\frac{1}{2}\right\}$

Now we have $2t - 1 - 3t\frac{4t-3}{4t-1} = \frac{2t(1-t)}{2t-1}$

Solving $t = \pm 1$ but $t \in (0,1) - \left\{\frac{1}{2}\right\}$

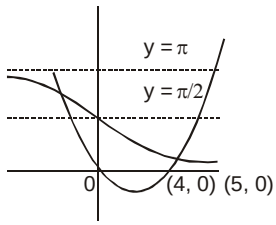
So there is no t, so there is no x



These positions can't be the answer

17. (P) Form the graph it is clear that x = 5 is the least positive integral value for which $x^2 - 4x > \cot^{-1} x$.

(Q) $f'(x) = 2e^x + ae^{-x} + 2a + 1$
$= e^{-x}(2e^{2x} + (2a+1)e^x + a)$
$= 2e^{-x}(e^{2x} + (a+1/2)e^x + a/2)$
$= 2e^{-x}(e^x + a)(e^x + 1/2)$
For f(x) to be increasing $f'(x) \geq 0\ \forall\ x \in R$
or $e^x + a \geq 0\ x \in R$
or $a \geq 0$



(R) $\lim_{x\to 0}\left(\frac{f^2(a+x)}{f(a)}\right)^{\frac{1}{x}} = e^{\lim_{x\to 0}\left(\frac{f^2(a+x)-f(a)}{xf(a)}\right)} = e^{\lim_{x\to 0}\frac{f^2(a+x)-(f(a))^2}{x}} = e^4$

$\therefore k = 4$

(S) $\sin^{-1}(-x^2 + 4x - 3) = \cos^{-1}(2 - |x|)$
$0 \leq -x^2 + 4x - 3 \leq 1 \Rightarrow x \in [1, 3]$ ..... (1)
Also, $0 \leq 2 - |x| \leq 1 \Rightarrow x \in [-2, -1] \cup [1, 2]$ ..... (2)
From (1) and (2) $x \in \{1, 2\}$

18. (P) Equation of any normal to $y^2 = 4x$ is $y = -tx + 2t + t^3$
Comparing it with $y = 2x + c - 4$
$t = -2$ and $2t + t^3 = c - 4$
$\therefore c = 4 - 4 - 8 = -8$
$\therefore -c = 8$

(Q) Equation of the chord whose mid point is (0, 3) is $\frac{3y}{25} - 1 = \frac{9}{25} - 1$

i.e. $y = 3$ it intersects the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ at $\frac{x^2}{16} = 1 - \frac{9}{25} = \frac{16}{25}$

$x = \pm \frac{16}{5}$

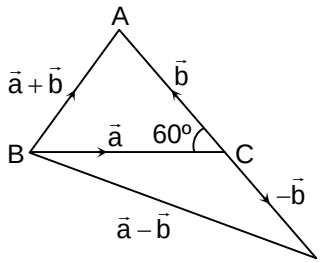
$\therefore$ Length of the chord $= \frac{32}{5}$

Thus $\frac{4k}{5} = \frac{32}{5}$ $\therefore k = 8$

(R) The product of the lengths of the perpendiculars from the two focii on any tangent to the hyperbola $\frac{x^2}{25} - \frac{y^2}{3} = 1$ is 3
$\therefore 3 = \sqrt{k}$, hence $k = 9$

(S) Since, $x + y = a$ touches the hyperbola
$x^2 - 2y^2 = 18$
$\therefore x^2 - 2(a - x)^2 = 18$ has equal roots
i.e., $x^2 - 4ax + 18 + 2a^2 = 0$ has equal roots
$\therefore 16a^2 - 4(18 + 2a^2) = 0$
$8a^2 - 72 = 0 \Rightarrow a = \pm 3$
$\therefore |b| = 3$

19. (P) From figure $\vec{a} \wedge \vec{b} = \frac{2\pi}{3}$



and $|\vec{a} - \vec{b}|^2 = 2^2 - 1$
$|\vec{a} - \vec{b}| = \sqrt{3} = \sqrt{x}$
$\therefore x = 3$

(Q) $f(x) = \sin x$, $f(x) + f(\pi - x) = 2$
$f(x) = 2 - f(\pi - x) = 2 - \sin(\pi - x) = 2 - \sin x$, where $x \in \left(\frac{\pi}{2}, \pi\right]$
$f(x) = f(2\pi - x) = 2 - \sin(2\pi - x)$, where $x \in \left(\frac{3\pi}{2}, 2\pi\right]$

$$f(x) = \begin{cases} \sin x & , \ x \in \left[0, \frac{\pi}{2}\right] \\ 2 - \sin x & , \ x \in \left(\frac{\pi}{2}, \pi\right] \\ 2 + \sin x & , \ x \in \left(\pi, \frac{3\pi}{2}\right] \\ -\sin x & , \ x \in \left(\frac{3\pi}{2}, 2\pi\right] \end{cases}$$

$$\text{Area} = \int_0^{\pi/2} \sin x\,dx + \int_{\pi/2}^{\pi} 2 - \sin x\,dx + \int_{\pi}^{3\pi/2} (2+\sin x)\,dx + \int_{3\pi/2}^{2\pi} (-\sin x)\,dx$$

$$= 1 + 2\times\frac{\pi}{2} - 1 + 2\cdot\frac{\pi}{2} - 1 + 1 = 2\pi \text{ sq. units}$$

(R) Since $g = f^{-1}$ so $fog(x) = x$
Therefore $f'(g(x))\, g'(x) = 1$
$f'(g(a))\, g'(a) = 1$

$\Rightarrow f'(b)\, 2 = 1 \Rightarrow f'(b) = \frac{1}{2}$

(S) Putting $x = \tan\theta$

$$\frac{\sqrt{1+x^2}-1}{x} = \frac{\sec\theta - 1}{\tan\theta} = \tan\frac{\theta}{2}$$

This $u = \tan^{-1}\frac{\sqrt{1+x^2}-1}{x} = \frac{\theta}{2} = \frac{1}{2}\tan^{-1}x$

Putting $x = \sin\theta$, $v = \tan^{-1}\frac{2x\sqrt{1-x^2}}{1-2x^2} = 2\sin^{-1}x$

$$\frac{du}{dv} = \frac{\frac{1}{2}\frac{1}{1+x^2}}{\frac{2}{\sqrt{1-x^2}}} \Rightarrow \left.\frac{du}{dv}\right]_{x=0} = \frac{1}{4}$$

20. (P) We have to select 3 no. from {1, 2, ...100}
Number division by 6 are 6, 12 96

$$\text{Prob} = \frac{{}^{16}C_3}{{}^{100}C_3} = \frac{4}{1155}$$

(Q) 22nd century 2200, 2201 2299
2200 leap year (L), 2201 Non leap year (N)

In 22nd century: 25 L and 75 N, $P(L) = \frac{1}{4}$ $P(N) = \frac{3}{4}$

For L: 366 days; $\frac{366}{7} = 52 + \frac{2}{7}$

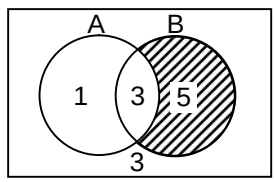
For N: 365 days; $\frac{365}{7} = 52 + \frac{1}{7}$

Now P(53 Sunday) = P(L) P(53S/L) + P(N)·P(53 Sun/N) $= \frac{1}{4}\times\frac{2}{7} + \frac{3}{4}\times\frac{1}{7} = \frac{5}{28}$

(R) P(at least one head) = 1 – P (no head) = $1 - {}^5C_5\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^0 = 1 - \frac{1}{32} = \frac{31}{32}$

(S) $P(A\cup B) = \frac{3}{4} = \frac{9}{12}$, $P(A\cap B) = \frac{1}{4} = \frac{3}{12}$

$P(\bar{A}) = \frac{8}{12}$; $P(A) = \frac{4}{12}$, $P(\bar{A}\cap B) = \frac{5}{12}$



S(12)